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Adaptive resource allocation in dynamic wireless environments: a systematic review

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Abstract

Increasing heterogeneity, densification, mobility, and service diversification have rendered traditional static resource allocation strategies ineffective for contemporary wireless networks. The transition to 5G-Advanced and emerging 6G systems further introduces highly dynamic environments characterised by continuously evolving traffic demand, interference patterns, network topology, and application requirements. This study presents a PRISMA-based systematic review of adaptive resource allocation in dynamic wireless environments, synthesising evidence from 34 peer-reviewed studies published from 2020 onward across major databases including IEEE Xplore, Scopus, Web of Science, ACM Digital Library, ScienceDirect, and SpringerLink. Unlike prior surveys, this review provides a unified cross-domain perspective by integrating modelling approaches, allocation strategies, and architectural contexts across terrestrial, edge-enabled, non-terrestrial, and semantic communication systems. It further develops a structured taxonomy of adaptive resource allocation frameworks, encompassing optimisation-based, learning-based, and hybrid methods under non-stationary conditions. The findings reveal a strong shift toward explicit dynamic modelling and the dominance of learning-based approaches, particularly deep reinforcement learning and multi-agent reinforcement learning, due to their ability to handle uncertainty and sequential decision-making. However, persistent challenges related to training stability, generalisation, scalability, and deployment safety are driving the emergence of hybrid and constraint-aware frameworks. The review also highlights the increasing integration of adaptive allocation within advanced architectures such as Open RAN, multi-access edge computing, non-terrestrial networks, and digital-twin-assisted systems. Despite these advances, evaluation practices remain largely simulation-driven, limiting reproducibility and real-world applicability. This study uniquely identifies critical methodological gaps and outlines deployment-oriented research directions, including cross-domain optimisation, trustworthy AI integration, and reproducible validation frameworks, to support the development of reliable and scalable adaptive resource allocation mechanisms for future 6G systems.

Keywords Adaptive resource allocation, Dynamic wireless networks, 6G communications, Learning-based optimization, Network architecture

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Introduction

The traditional static or quasi-static resource planning approaches have become less effective to control the next-generation wireless systems, where traffic demand, user mobility, interference conditions, and application requirements vary at fine temporal and spatial granularities over highly heterogeneous infrastructures [17, 29]. Especially in dense 5G HetNets, when static allocation policies are used, the coexistence of macro cells, small cells, and diverse access technologies would cause rapidly changing interference patterns and uneven traffic distributions, resulting in persistent load imbalance and resource underutilisation [2]. On top of that, supporting multiple service classes (e.g., enhanced Mobile Broadband (eMBB), Ultra-Reliable Low-Latency Communications (URLLC), and massive Internet of Things (mIoT)) with distinct and often conflicting quality-of-service (QoS) requirements makes resource allocation no longer a single-objective throughput maximisation problem, but a multi-dimensional trade-off problem with respect to latency, reliability, fairness, and energy efficiency. Mathematically, adaptive resource allocation can be formulated as a time-varying constrained optimisation problem. Let (*t*) represent the vector of allocatable resources at time *t*, such as spectrum blocks, transmit power, scheduling intervals, computing cycles, or network slices. The adaptive allocation objective can be written as

$$_{x(t)}{}^{max} U\left(x\left(t\right), s\left(t\right)\right) \tag{1}$$

Subjected to $x\left(t\right) \in \mathrm{X}, g_k\left(X\left(t\right), s\left(t\right)\right) \leq \forall \mathrm{k} \in \{1, \ldots, \mathrm{K}\}$

where $x\left(t\right)$ denotes the vector of resource allocation decisions at time t (e.g., spectrum allocation, transmit power, scheduling, or computing resources); X denotes the feasible allocation set defined by system constraints; $s(t)$ denotes the network state at time t (e.g., traffic load, channel conditions, interference levels, and mobility); $U(x(t), s(t))$ denotes the system utility function capturing performance objectives such as throughput, latency, fairness, and energy efficiency; $g_k\left(X\left(t\right), s\left(t\right)\right)$ denotes the $k-th$ constraint function representing system limitations such as QoS requirements, power bounds, latency constraints, and fairness conditions; and K denotes the total number of constraints.

$$x\left(t\right) \in X, g\left(x\left(t\right), s\left(t\right)\right) \leq 0, \tag{2}$$

where $U(\cdot)$ denotes the system utility and X represents feasibility constraints imposed by physical, regulatory, and architectural limits. The constraint function $g(\cdot)$ captures quality-of-service, latency, reliability, power, and fairness requirements. In multi-service environments, the utility is inherently multi-objective and is commonly represented as a weighted combination of competing performance metrics,

$$U = \alpha R - \beta D - \gamma E, \tag{3}$$

where R is achievable throughput, D denotes end-to-end latency, E represents energy consumption, and α, β, and γ are service-dependent weights reflecting the priorities of eMBB, URLLC, and mIoT traffic. This formulation emphasises the inherent trade-offs that adaptive resource allocation mechanisms have to address under dynamic network conditions, and the further complexity that the transition to sixth-generation (6G) wireless systems with fundamentally new operational paradigms brings with it,

such as end-to-end orchestration of resource allocation across the RAN, transport network, core network, and distributed computing resources [7, 31], pervasive virtualisation, tighter edge–cloud integration, and use of a variety of spectrum bands including sub-6 GHz, millimetre-wave, and terahertz frequencies [7, 31]. In these environments, network states are constantly changing as users move, services are instantiated or terminated, and network functions scale elastically, which violates the assumptions of temporal stability that underpin many classical optimisation and control techniques [3, 23]. Instead, they need to learn, adapt, and generalise under uncertainty, rather than rely on fixed or slowly updated analytical models.

As illustrated in Fig. 1, the conceptual framework of adaptive resource allocation highlights the interaction between dynamic network states, allocation decisions, and system performance objectives in modern wireless environments.

Learning-based approaches particularly deep reinforcement learning (DRL) have therefore emerged as prominent tools for adaptive resource allocation in dynamic wireless environments. DRL is well suited to sequential decision-making problems where system dynamics are partially observable and rewards are delayed, making it attractive for applications such as network slicing, admission control, and real-time scheduling. The suitability of deep reinforcement learning for adaptive resource allocation can be formalised by modelling the problem as a Markov Decision Process (MDP) defined by the tuple

$$M = (S, A, P, R, \gamma)\,, \tag{4}$$

where S denotes the space of network states, A represents the set of possible allocation actions, P characterises state transition probabilities, R is a reward function encoding



Fig. 1 Conceptual framework of adaptive resource allocation in dynamic wireless networks

QoS and efficiency objectives, and $\gamma \in (0,1)$ is a discount factor. The goal is to learn an allocation policy π that maximises the expected long-term cumulative reward,

$$\max_{\pi} E_{\pi}\left[\sum_{t=0}^{\infty} \gamma^{t} r\left(t\right)\right] \tag{5}$$

This formulation captures the sequential, time-coupled nature of adaptive resource allocation, where present decisions influence future network states and performance outcomes, thereby justifying the growing adoption of learning-based optimisation in dynamic wireless environments.

Recent research shows that DRL-based frameworks can adjust resource allocation policies dynamically in accordance with time-varying traffic loads and SLA constraints and can outperform static or rule-based schemes in highly dynamic network scenarios [12, 25]. However, the literature also points out several practical challenges that limit the deployment of DRL in operational networks, including the exploration overhead, sample inefficiency, instability under distributional shifts, and the risk of transient QoS degradation during learning phases [23, 33]. The evolution of open and disaggregated radio access network architectures further changes the landscape of adaptive resource allocation, where the Open RAN (O-RAN) framework defines programmable control interfaces (the near-real-time and non-real-time RAN Intelligent Controllers (RICs)) that make it possible to implement resource allocation logic as data-driven xApps and rApps, which can enable greater flexibility to deploy adaptive, AI-assisted allocation strategies based on real-time performance measurements and high-level policy objectives [22]. However, more openness also presents challenges, including multi-vendor interoperability, policy conflicts among independently developed applications, and an expanded attack surface that require resource allocation mechanisms that are not only adaptive but also robust, secure, and compliant with evolving O-RAN specifications [21, 24].

In parallel, the increasing reliance on multi-access edge computing (MEC) to satisfy stringent latency and reliability requirements further blurs the traditional boundary between communication and computation. In dynamic wireless environments, radio resource allocation decisions directly affect edge workload distribution, while the availability of computing resources constrains achievable communication performance. This strong interdependence transforms adaptive resource allocation into a cross-domain optimisation problem involving radio spectrum, transmit power, computing cycles, and storage capacity [13, 14].

$$D_{total} = D_{radio} + D_{compute} \leq D_{max} \tag{6}$$

where D_{radio} denotes wireless transmission delay, $D_{compute}$ represents edge or cloud processing delay, and D_{max} is the application-specific latency bound. This relationship illustrates that radio-only optimisation is insufficient in MEC-enabled environments, as communication and computation resources must be jointly allocated to satisfy stringent QoE requirements under dynamic load and mobility conditions. Approaches that treat these resources independently risk suboptimal performance or instability, particularly under bursty traffic and mobility-driven demand shifts. Consequently, coordinated radio–compute–storage allocation strategies are increasingly recognised as essential for sustaining quality of experience (QoE) in latency-sensitive and compute-intensive applications. Against this backdrop, a systematic review of adaptive resource allocation

in dynamic wireless environments is both timely and necessary. While a growing body of work addresses individual aspects such as optimisation-based scheduling, learning-driven control, or architectural enablers the literature remains fragmented across layers, technologies, and methodological perspectives. There is a clear need to consolidate how adaptive resource allocation problems are formulated, which solution families dominate under different modelling assumptions, how performance and robustness are evaluated, and where critical gaps persist with respect to scalability, reliability, and real-world deployability [3, 29]. By synthesising recent advances and systematically identifying unresolved challenges, such a review can provide a coherent foundation for future research aimed at designing dependable, efficient, and adaptive wireless networks capable of operating under sustained dynamics and uncertainty.

This study makes four key contributions to the field of adaptive resource allocation in dynamic wireless environments. First, it provides a PRISMA-based systematic synthesis of recent literature (2020–2025), capturing the shift toward dynamic and non-stationary network modelling. Second, it develops a unified cross-domain perspective by integrating insights from 5G/6G networks, multi-access edge computing, non-terrestrial networks, and semantic communication systems. Third, the study proposes a structured taxonomy of modelling approaches and allocation strategies, encompassing learning-based, optimisation-based, and hybrid frameworks under realistic dynamic conditions. Finally, it identifies critical methodological and evaluation gaps, particularly the dominance of simulation-based validation, and outlines directions for more reliable and deployment-oriented research.

The remainder of this paper is organised as follows. Section II presents the problem statement. Section III reviews related studies. Section IV describes the methodology. Section V presents the results. Section VI discusses the findings. Section VII concludes the paper and outlines future research directions.

Problem statement

Despite extensive research on radio resource management, adaptive resource allocation in dynamic wireless environments remains fundamentally unresolved. Contemporary wireless systems operate under persistent non-stationarity caused by user mobility, bursty traffic arrivals, fluctuating interference, virtualization dynamics, and heterogeneous service requirements. Classical resource allocation models typically assume quasi-static network states and formulate optimisation problems over fixed or slowly varying parameters. However, in 5G-Advanced and emerging 6G networks, the network state evolves continuously over time, rendering static or offline optimisation ineffective. Formally, let the network state at time t be denoted by $s(t)$, comprising traffic load, channel conditions, interference levels, mobility patterns, and service demands. The state evolution follows unknown stochastic dynamics,

$$s\left(t+1\right)=f\left(s\left(t\right),x\left(t\right),\xi\left(\mathrm{t}\right)\right), \tag{7}$$

where $x(t)$ represents the resource allocation decision and $\xi(t)$ captures exogenous uncertainties such as traffic bursts and channel fading. Under such highly dynamic conditions, static or poorly adaptive resource allocation policies fail to maintain service guarantees, resulting in quality-of-service (QoS) degradation, service-level agreement (SLA) violations, and network instability, alongside inefficient resource

utilisation and increased operational costs. Although learning-based and architecture-aware approaches have been proposed to address these challenges, existing studies remain fragmented across modelling assumptions, algorithmic paradigms, architectural contexts, and evaluation practices. This fragmentation limits a coherent understanding of how adaptive resource allocation problems should be rigorously formulated under non-stationary conditions, which solution paradigms are most effective across varying levels of uncertainty, and how adaptability can be systematically balanced with stability, fairness, and scalability. Consequently, there is a critical need for a structured and comprehensive synthesis of current research to clarify these issues and guide the development of robust and deployable adaptive resource allocation frameworks for next-generation wireless systems.

Related studies

Recent scholarship on adaptive resource allocation in dynamic wireless environments reflects a decisive shift from static, centrally planned models toward distributed, learning-driven, and context-aware control mechanisms. This evolution is driven by increasing network heterogeneity, service differentiation, and environmental non-stationarity introduced by mobility, traffic bursts, virtualization, and multi-tenant operation. Accordingly, prior studies can be broadly categorized into four interrelated strands: (1) learning-based slicing and multi-service scheduling, (2) multi-agent coordination for decentralized resource control, (3) edge-assisted and cross-domain orchestration, and (iv) emerging 6G-oriented contexts such as semantic communications, vehicular networking, non-terrestrial networks, and digital-twin-enabled optimization. Collectively, these bodies of work move away from single-cell, stationary assumptions and instead emphasize multi-objective decision-making under uncertainty, highlighting the need for adaptive mechanisms capable of operating across layers, domains, and administrative boundaries.

Learning-Based slicing and multi-service scheduling in dynamic networks

Recent research shows that DRL-based frameworks can adjust resource allocation policies dynamically in accordance with time-varying traffic loads and SLA constraints and can outperform static or rule-based schemes in highly dynamic network scenarios [12, 25]. However, the literature also points out several practical challenges that limit the deployment of DRL in operational networks, including the exploration overhead, sample inefficiency, instability under distributional shifts, and the risk of transient QoS degradation during learning phases [23, 33]. The evolution of open and disaggregated radio access network architectures further changes the landscape of adaptive resource allocation, where the Open RAN (O-RAN) framework defines programmable control interfaces (the near-real-time and non-real-time RAN Intelligent Controllers (RICs)) that make it possible to implement resource allocation logic as data-driven xApps and rApps, which can enable greater flexibility to deploy adaptive, AI-assisted allocation strategies based on real-time performance measurements and high-level policy objectives [22]. However, more openness also presents challenges, including multi-vendor interoperability, policy conflicts among independently developed applications, and an expanded attack surface that require resource allocation mechanisms that are not only adaptive but also robust, secure, and compliant with evolving O-RAN specifications [21, 24].

The literature demonstrates a clear trend toward deep reinforcement learning–based approaches for adaptive slicing and multi-service scheduling, primarily due to their ability to capture time-varying traffic dynamics and QoS requirements. However, these approaches are constrained by training instability, sample inefficiency, and vulnerability to distributional shifts, which limit their reliability in real-world deployments. This reveals a critical research gap in developing stable, sample-efficient, and deployment-ready learning frameworks that can operate reliably under highly dynamic wireless conditions.

Distributed and multi-agent reinforcement learning for resource allocation

As wireless networks scale in density, coverage, and administrative complexity, many resource allocation decisions become inherently decentralized. Inter-cell interference management, multi-tenant slicing, vehicular coordination, and multi-RIC control in O-RAN environments cannot be efficiently addressed by single-agent controllers with global state visibility. In response, recent studies increasingly adopt multi-agent reinforcement learning (MARL) formulations, where multiple learning agents cooperate or compete while making localized decisions based on partial observations. A multi-agent wireless resource allocation problem can be modeled as a Decentralized Partially Observable Markov Decision Process (Dec-POMDP):

$$M = \langle S, \{A_i\}_{i=1,}^{N} \{O_i\}_{i=1,}^{N} P, R, \gamma \rangle \tag{8}$$

where M denotes the Decentralized Partially Observable Markov Decision Process; S denotes the global state space of the wireless network, including quantities such as traffic load, interference conditions, channel states, and mobility patterns; A_i denotes the action space of agent iii, which may include decisions such as transmit power selection, spectrum allocation, scheduling choice, or slice adjustment; N denotes the total number of agents participating in the distributed allocation process; O_i denotes the observation space of agent i, representing the partial and local information available to that agent; P(s′|s,a1,...,a_N) denotes the state transition probability function, which specifies the probability of moving from state s to state $s\prime$ after the joint action $(a_1, \ldots, a_N)$ is taken; R(s,$a_1, \ldots, a_N$) denotes the reward function, which may be defined as a shared global reward or as individual agent rewards depending on the coordination setting; and $\gamma \in (0,1)$ denotes the discount factor, which determines the relative importance of future rewards in sequential decision-making.

A notable example is a multi-agent reinforcement learning (MARL)-based framework developed for Internet of Vehicles (IoV) scenarios, which explicitly targets distributed execution under high mobility and rapidly changing network topologies. By enabling agents to learn coordinated policies while operating independently, such frameworks demonstrate improved scalability and responsiveness compared to centralized approaches [27]. Complementing these application-specific studies, comprehensive MARL surveys synthesise recent advances in coordination mechanisms, credit assignment strategies, and convergence guarantees. These surveys consistently emphasise that scalability, stability, and communication overhead remain critical bottlenecks for real-time wireless deployment, particularly in dense and fast-varying environments [19]. At lower protocol layers, reinforcement learning has also been applied to adaptive contention and traffic control, illustrating that resource allocation extends beyond high-level

RAN scheduling. For instance, deep reinforcement learning-based contention window optimisation has been shown to reduce collisions and improve efficiency under time-varying traffic loads, reinforcing the perspective that adaptive allocation must span MAC-to-RAN layers in modern wireless systems [26]. Collectively, these studies underscore the growing importance of agent-based and distributed learning paradigms for managing complexity in highly dynamic wireless environments. The literature indicates a strong shift toward multi-agent reinforcement learning as a scalable solution for decentralised resource allocation in complex wireless environments. While MARL improves scalability and responsiveness under partial observability, it is limited by coordination complexity, communication overhead, and convergence challenges. This highlights a key research gap in designing efficient coordination and learning mechanisms that can balance scalability with stability and low signalling overhead in distributed wireless systems.

Edge-assisted orchestration and compute-aware resource allocation

The proliferation of latency-sensitive and compute-intensive applications has intensified the coupling between wireless access and multi-access edge computing (MEC). In such environments, adaptive resource allocation must jointly consider radio conditions, computational capacity, task arrival dynamics, and service deadlines. Recent surveys on resource scheduling in MEC environments synthesize a wide range of approaches that integrate offloading decisions, task placement, and radio resource allocation under stochastic arrivals and stringent latency constraints [13]. These studies demonstrate that treating communication and computation resources independently often results in suboptimal performance or even system instability. This cross-domain perspective is further reinforced by research on networked intelligence, where resource allocation decisions are tightly linked to machine learning workloads executed at the edge. In these scenarios, allocation policies must account for training and inference requirements, data availability, and model update frequencies, motivating compute-aware and learning-aware scheduling strategies rather than purely link-centric optimisation [19]. Collectively, these findings highlight a broader trend toward holistic orchestration frameworks, in which adaptive allocation spans radio, compute, and storage resources particularly under dynamic and unpredictable operating conditions. A major trend in the literature is the increasing integration of communication and computation resource allocation within edge-enabled environments, reflecting the need for cross-domain optimisation. However, many existing approaches rely on simplified assumptions or treat radio and compute resources independently, limiting their effectiveness in realistic deployments. This exposes a significant research gap in developing unified, cross-domain allocation frameworks that jointly optimise communication, computation, and storage resources under dynamic and heterogeneous conditions.

Emerging 6G contexts shaping adaptive allocation research

These emerging 6G networks introduce fundamentally new operational contexts that will significantly reshape adaptive resource allocation research: 6G is characterised by extreme heterogeneity, AI-native control, tight integration of communication, sensing, and computing, and highly dynamic network topologies; semantic communications shifts allocation objectives from bit-level accuracy toward task relevance and meaning fidelity, non-terrestrial networks introduce rapid topology changes, long-delay

links, and intermittent connectivity that challenge conventional scheduling and slicing mechanisms; Open RAN and deep virtualization expose programmable control points that enable real-time, data-driven allocation but also raise concerns related to security, policy conflicts, and governance; and digital twin–assisted control further enables risk-aware and closed-loop optimisation by allowing policies to be evaluated under realistic dynamics before deployment. Emerging 6G research highlights a paradigm shift toward context-aware, AI-driven, and highly dynamic resource allocation frameworks spanning semantic communication, non-terrestrial networks, and digital twin–enabled systems. While these approaches offer significant potential for improving adaptability and efficiency, they remain largely conceptual or simulation-driven, with limited real-world validation and standardisation. This underscores a critical research gap in translating these advanced concepts into scalable, interoperable, and experimentally validated solutions suitable for deployment in next-generation wireless networks.

Semantic communications and meaning-aware optimization

Semantic communications introduce a fundamental paradigm shift by redefining communication efficiency in terms of task success, semantic relevance, or inference accuracy, rather than strict bit-level fidelity. This reconceptualisation challenges long-standing assumptions underlying classical resource allocation, which traditionally optimises throughput, spectral efficiency, or packet error rates. In semantic-aware systems, allocation decisions must instead account for the contextual importance of transmitted information, the tolerance of applications to distortion, and the interaction between communication and downstream learning or decision processes. Consequently, resource allocation objectives are increasingly framed around relevance-aware scheduling, adaptive compression, and task-oriented prioritisation rather than uniform rate maximisation. Recent tutorials and surveys argue that semantics-empowered networking will require new performance metrics, control abstractions, and cross-layer optimisation strategies, as conventional rate-based metrics may fail to capture application-level utility in semantic tasks [28, 30]. Domain-specific studies, particularly in Internet of Vehicles (IoV) environments, further demonstrate that semantic architectures are tightly coupled with mobility, perception tasks, and real-time decision-making, making adaptive resource allocation central to sustaining performance under rapidly changing conditions [30]. At the broader system level, semantic-aware resource allocation is increasingly positioned as a cornerstone of intelligent wireless ecosystems envisioned for 6G, where communication, sensing, and learning are co-designed to support context-aware and goal-oriented services [34].

Non-terrestrial networks and highly dynamic topology

Networks, such as low-Earth-orbit (LEO) satellite constellations and high-altitude platform systems, which are also referred to as non-terrestrial networks (NTNs), significantly complicate adaptive resource allocation because the network topology, propagation delay, and link intermittency are significantly different from that of conventional terrestrial cellular networks. NTN links experience frequent handovers, long and variable round-trip times, and rapidly changing visibility constraints, which invalidate many assumptions underlying conventional scheduling, access control, and slicing mechanisms. NTNs significantly complicate adaptive resource allocation, as decisions

must consider both spatio-temporal mobility and heterogeneous latency regimes across integrated terrestrial–non-terrestrial architectures. AI-empowered NTN networking is recognized as a critical research frontier in a recent survey in IEEE Communications Surveys & Tutorials [18], which underscores the need for adaptive resource allocation and mobility management to ensure acceptable quality of service and reliability in integrated 6G systems and identifies learning-based and predictive approaches as particularly promising for handling NTN dynamics, while also identifying open challenges in scalability, coordination with terrestrial RANs, and robustness under sparse or delayed feedback. Together, these results position NTNs as a promising testbed for advancing adaptive allocation strategies capable of operating under extreme and rapidly evolving network conditions.

Digital twins for closed-loop, risk-reduced optimization

In recent IEEE Communications Magazine studies, DT technologies are increasingly recognized as an important enabler for the safe and effective deployment of adaptive and learning-based resource allocation mechanisms [4], in which the high-fidelity, real-time replicas of network behaviour allow candidate allocation policies to be evaluated under realistic traffic, mobility, and interference dynamics before being applied to operational infrastructure, which is particularly valuable for learning-based approaches where uncontrolled exploration or policy instability can degrade quality of service if directly deployed in live networks. In addition to offline testing, DTs are also expected to serve as online companions for continuous monitoring and what-if analysis, predictive control, and rapid policy adaptation [4]. Therefore, DT-assisted allocation frameworks are a promising direction for the integration of adaptability and operational reliability in future wireless networks.

Practical O-RAN control-loop concerns: security, governance, and policy conflicts

The adoption of O-RAN architectures embeds intelligence directly into near-real-time control loops, fundamentally reshaping how adaptive resource allocation is designed and deployed. While programmability through xApps and rApps enables fine-grained, data-driven optimization, it also raises critical concerns related to trust, security, and governance. In open, multi-vendor ecosystems, resource allocation decisions may be influenced by independently developed applications with differing objectives, increasing the risk of policy conflicts, unstable control behavior, or unintended performance degradation. Recent studies argue that adaptive optimization frameworks must therefore explicitly incorporate security and accountability considerations, including xApp integrity verification, protection against model manipulation and data poisoning, and continuous runtime monitoring of control actions [9]. From this perspective, adaptive resource allocation can no longer be treated as a purely performance-driven problem. Instead, it must be governed by policy-aware and trust-aware mechanisms that ensure predictable, fair, and secure operation across stakeholders. These concerns underscore the need for holistic allocation frameworks that jointly address performance, robustness, and governance in O-RAN-based 6G systems.

Methodology

This study adopted a systematic review methodology to identify, evaluate, and synthesise recent scholarly work on adaptive resource allocation in dynamic wireless environments. A systematic approach was chosen to ensure transparency, reproducibility, and comprehensive coverage of the rapidly evolving wireless networking literature, particularly in the context of 5G and emerging 6G systems. Given the interdisciplinary nature of adaptive resource allocation spanning radio access networks, edge computing, learning-based control, and emerging architectural paradigms the review was designed to integrate evidence across multiple research communities while maintaining methodological rigor. The review process was guided by established systematic review principles commonly used in engineering and computer science research, with procedures aligned to PRISMA recommendations where applicable, while remaining compatible with IEEE-style survey expectations as illustrated in Fig. 2.

Review protocol and design

The review protocol was defined prior to the literature search to minimise selection bias and ensure consistency throughout the study. The protocol specified the review objectives, research scope, target publication period, databases to be searched, and criteria for study selection and quality assessment. The review focused on peer-reviewed journal

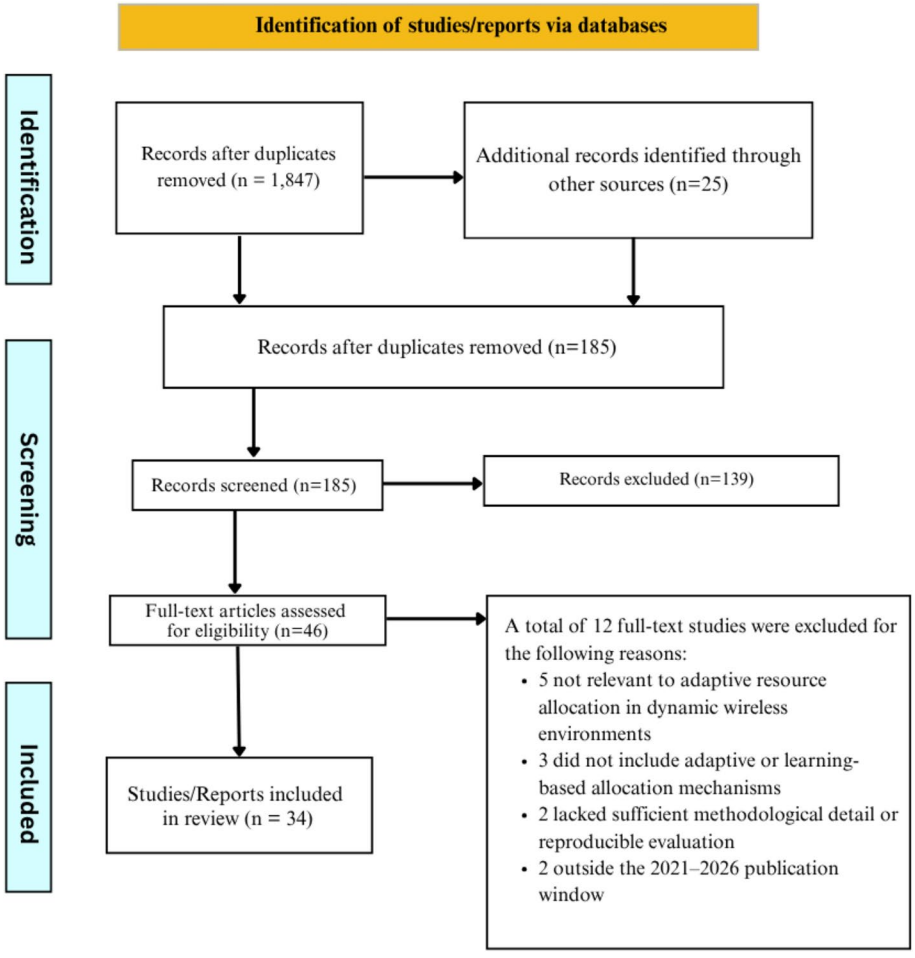


Fig. 2 PRISMA flow diagram of the study selection process

articles and conference papers published from 2020 onward, reflecting the period in which adaptive, learning-driven, and software-defined wireless networking became central research themes. The protocol also established the thematic dimensions used for synthesis, including problem formulations, solution approaches, architectural contexts, and evaluation practices, enabling structured comparison across studies.

Research questions

This systematic review is guided by several key research questions. Specifically, it examines how adaptive resource allocation problems should be formulated under non-stationary wireless network conditions. It further investigates which solution paradigms are most effective under varying levels of uncertainty. In addition, the study explores how adaptability can be balanced with critical system requirements such as stability, fairness, and scalability. Finally, it evaluates the limitations of current validation and evaluation methodologies used in adaptive resource allocation research.

Data sources and search strategy

The literature search was conducted across multiple major scientific databases to ensure comprehensive coverage of high-quality research in wireless communications and networking. These databases included IEEE Xplore, Scopus, Web of Science, ACM Digital Library, ScienceDirect, and SpringerLink. A structured search strategy was employed using combinations of keywords related to adaptive resource allocation, dynamic wireless environments, learning-based optimisation, network slicing, edge computing, O-RAN, and 6G systems. Boolean operators and synonym expansion were applied to capture variations in terminology across disciplines. The search was restricted to English-language publications to ensure consistency in analysis and interpretation.

Study selection process

The study selection process followed a multi-stage screening procedure to progressively refine the initial search results into a final corpus of relevant studies. First, duplicate records were removed across databases. Second, titles and abstracts were screened to exclude studies that were clearly outside the scope of adaptive resource allocation in wireless networks. Third, full-text screening was conducted to assess methodological relevance, technical depth, and alignment with the review objectives. At each stage, studies were retained only if they explicitly addressed adaptive or dynamic resource allocation under time-varying network conditions. This staged approach ensured both breadth and relevance while reducing the likelihood of including marginal or tangential contributions.

Inclusion and exclusion criteria

Inclusion criteria

i. Studies that focus on adaptive or dynamic resource allocation in wireless communication or networking systems
ii. Research addressing 5G, 5G-Advanced, 6G, or beyond-5G network environments
iii. Articles proposing or analysing optimisation-based, learning-based, or hybrid resource allocation approaches
iv. Peer-reviewed journal articles and conference papers published from 2020 onward

v. Studies providing quantitative or qualitative performance evaluation

Exclusion criteria

i. Studies limited to static or offline resource planning without adaptive mechanisms
ii. Papers focused solely on non-wireless domains or wired networking
iii. Articles lacking sufficient methodological detail or evaluation results
iv. Editorials, tutorials without original synthesis, theses, and non-peer-reviewed reports
v. Publications not written in English

Quality assessment of selected studies

To ensure the reliability and relevance of the included studies, a quality assessment procedure was applied during full-text review. Each study was evaluated based on clarity of problem formulation, appropriateness of the proposed methodology, transparency of assumptions, adequacy of performance evaluation, and discussion of limitations. Studies that lacked baseline comparisons, relied on unrealistic assumptions, or provided insufficient experimental detail were scrutinised carefully to determine their contribution to the synthesis. This assessment did not serve to exclude studies solely on quality grounds but informed the weighting and interpretation of evidence during thematic analysis.

Data extraction process

A structured data extraction framework was used to systematically collect relevant information from each selected study. Extracted data included publication details, network scenario assumptions, types of resources considered, objective functions, algorithmic approaches, architectural context, and evaluation metrics. Additional information regarding datasets, simulators, or testbeds used for validation was also recorded. This structured extraction enabled consistent comparison across studies and facilitated the identification of trends, methodological clusters, and recurring limitations.

Data synthesis and analysis

The synthesis of findings was conducted using a narrative and thematic analysis approach. Rather than aggregating results through statistical meta-analysis which is often impractical due to heterogeneous assumptions and metrics in wireless networking research the review organised studies into conceptual themes reflecting dominant problem settings and solution strategies. Comparative tables were used to highlight similarities and differences across approaches, while qualitative synthesis was employed to interpret methodological strengths, weaknesses, and gaps. This approach allowed the review to capture both technical detail and broader research trajectories within adaptive resource allocation.

Threats to validity

Several potential threats to validity were considered in this review. Publication bias may arise from the exclusion of unpublished or non-peer-reviewed work, while language bias may result from restricting the review to English-language studies. Additionally, the rapid evolution of wireless technologies means that some recent developments may not yet be fully reflected in the literature. These limitations were mitigated by using multiple databases, broad search terms, and a clearly defined selection protocol. Nonetheless, the

findings should be interpreted in light of these constraints, particularly when generalising results to future 6G deployments.

Results

In this section, we synthesise the findings of the systematic review into a comprehensive report based on the final set of studies that passed the inclusion and exclusion criteria based on the PRISMA-guided selection process. We organise the results according to the study objectives to provide a structured and transparent account of how adaptive resource allocation has been investigated in dynamic wireless environments. Instead of presenting isolated outcomes from individual studies, this Results section integrates evidence across the literature to identify dominant modelling assumptions, methodological trends, and architectural contexts shaping current research. The results are presented in two complementary forms: summary tables that provide a comparative overview of representative studies and an interpretative narrative that analyses the aggregated findings, contextualises them within broader research trends, and draws out their implications for the design and deployment of adaptive resource allocation mechanisms. The presentation highlights how dynamic modelling, learning-driven optimisation, and evolving network architectures collectively influence adaptive resource allocation strategies, while also exposing methodological and practical limitations that inform subsequent discussion and identification of research gaps.

The results presented in Table 1 reveal a clear and progressive shift in how wireless environments are conceptualised within adaptive resource allocation research. Early studies in 5G heterogeneous networks primarily relied on stochastic representations

Table 1 To examine how dynamic wireless environments are modelled in adaptive resource allocation studies

Author(s) and Year	Network context	Dynamics type	Modelling strategy	Adaptivity level	Realism level	Key strength	Limitation
Manap et al. (2020)	5G HetNets	Traffic & interference	Stochastic models	Low	Medium	Simple baseline modelling	Fails under high dynamics
Xu et al. (2021)	Multi-tier 5G	Mobility & load	Time-varying states	Medium	Medium	Improved fairness	Limited scalability
Lee et al. (2020)	Wireless RAN	Uncertainty	State abstraction	Medium	Medium	Robust across scenarios	Lacks temporal depth
Chen et al. (2023)	Integrated HetNets	Demand fluctuation	DRL-based modelling	High	High	Adaptive slicing	Training complexity
Ssengonzi et al. (2022)	Virtualised 5G	Slice dynamics	Virtual state models	High	Medium	SLA-aware modelling	Limited real-world validation
Wang et al. (2025)	IoV	Mobility & topology	MARL-based	Very high	High	Scalable distributed control	Coordination overhead
Ye et al. (2025)	Semantic IoV	Task dynamics	Semantic state modelling	Very high	High	Reduces redundancy	Immature metrics
Mahboob et al. (2024)	NTN–6G	Satellite mobility	Predictive AI models	High	Medium	Handles extreme dynamics	Sparse feedback issues
Al-Hraishawi et al. (2024)	6G NTN	Multi-domain	Digital twin modelling	Very high	Very High	Safe evaluation	High complexity
Zhang (2023)	Semantic 6G	Contextual relevance	Meaning-aware states	Very high	High	Task efficiency	Lack of standardisation

of traffic and interference, which exposed the limitations of static allocation under dynamic conditions. More recent work increasingly incorporates user mobility, topology evolution, and service demand variability, providing a more realistic representation of operational environments. This progression is further extended in vehicular and semantic communication contexts, where modelling evolves toward task-level and context-aware dynamics, reflecting application-driven resource demands beyond traditional channel-based assumptions. In advanced scenarios such as non-terrestrial networks and 6G systems, extreme non-stationarity characterised by satellite mobility and intermittent connectivity fundamentally challenges conventional modelling approaches. A key emerging trend is the adoption of digital twin–based models, which enable multi-domain representation across communication, computation, and control layers while supporting risk-aware experimentation. However, despite these advancements, many models remain constrained by high complexity, limited scalability, and insufficient real-world validation. This highlights a critical research gap in developing scalable, generalisable, and experimentally validated dynamic modelling frameworks. The findings indicate that non-stationarity is now treated as a fundamental design condition rather than an exception, necessitating adaptive resource allocation strategies that can operate reliably under rapid and multi-dimensional environmental change.

The results presented in Table 2 indicate a clear dominance of learning-based approaches in adaptive resource allocation, with deep reinforcement learning (DRL) emerging as the most widely adopted technique due to its ability to handle sequential decision-making, uncertainty, and time-coupled constraints in dynamic wireless environments. Single-agent DRL methods are primarily applied to centralized scheduling and slicing problems, while multi-agent reinforcement learning (MARL) is increasingly utilised in distributed scenarios such as vehicular networks and multi-cell coordination, where scalability and partial observability are critical. A key trend is the growing transition from purely learning-driven approaches toward hybrid frameworks that integrate reinforcement learning with optimisation constraints, architectural controls such as O-RAN xApps, and federated mechanisms to address privacy, security, and coordination challenges. However, despite their adaptability, learning-based approaches remain limited by issues of training instability, lack of convergence guarantees, fairness concerns, and limited deployment readiness. Optimisation-based methods, although less dominant, continue to play an important role in providing interpretability and performance guarantees. These findings highlight a fundamental trade-off between adaptability and reliability, revealing a critical research gap in the development of scalable, stable, and trustworthy hybrid resource allocation frameworks that can balance learning flexibility with operational constraints.

The results of Table 3 reveal a clear shift in adaptive resource allocation from isolated radio access network optimisation toward integration within advanced architectural frameworks, including O-RAN, multi-access edge computing, non-terrestrial networks, semantic communication systems, and digital twin–enabled control loops. This reflects a broader trend toward cross-domain and architecture-aware resource allocation, where communication, computation, and control are jointly orchestrated to support dynamic and heterogeneous network environments. However, despite these architectural advancements, Table 3 highlights a persistent limitation in current evaluation practices, with the majority of studies relying heavily on simulation-based validation and lacking

Table 2 To identify dominant adaptive resource allocation approaches used in dynamic wireless environments

Author(s) and Year	Approach type	Resources optimised	Algorithmic technique	Adaptivity level	Scalability	Deployment readiness	Key strength	Limitation
Lee et al. (2020)	Learning-based	Spectrum, power	DRL	High	Medium	Low	Generalisable policies	Training instability
Hurtado Sánchez et al. (2022)	Learning-based	Bandwidth	DRL slicing	High	Medium	Low	SLA-aware adaptation	Sample inefficiency
Chen et al. (2023)	Learning-based	Bandwidth	Dueling DQN	High	Medium	Low	Improved stability	Limited real-world validation
Ssengonzi et al. (2022)	Learning-based	Multi-resource	DRL survey	High	Medium	Low	Comprehensive insight	Lacks implementation
Wang et al. (2025)	Learning-based	Spectrum, routing	MARL	Very High	High	Medium	Distributed scalability	Coordination overhead
Tu & Ma (2025)	Learning-based	MAC contention	DRL	High	Medium	Low	Reduced collisions	Narrow scope
Omheni et al. (2024)	Hybrid	Spectrum, energy	RL + O-RAN xApp	Very High	High	High	Real-time adaptability	Integration complexity
El-Hajj (2025)	Hybrid	Multi-domain	FL + RL	Very High	High	Medium	Trust-aware optimisation	Communication overhead
Aslan et al. (2024)	Optimisation-based	Virtualised RAN	Fairness optimisation	Medium	High	High	Strong fairness guarantees	Limited adaptability
Ismail et al. (2025)	Hybrid	Radio–compute	Joint optimisation	Very High	High	Medium	Latency reduction	Model complexity

large-scale, real-world experimentation. Although digital twin frameworks offer promising capabilities for closed-loop and risk-aware testing, their adoption remains constrained by the absence of standardised benchmarks, reproducible methodologies, and validated deployment pipelines, particularly in highly dynamic 6G and non-terrestrial network scenarios. This imbalance between architectural sophistication and evaluation maturity exposes a critical research gap in the development of realistic, standardised, and reproducible validation frameworks that can bridge the gap between theoretical models and operational deployment.

Discussion

This section discusses the findings of the systematic review in relation to existing literature on adaptive resource allocation in wireless networks. Moving beyond descriptive synthesis, the discussion critically interprets how recent studies conceptualise dynamic wireless environments, the methodological paradigms that dominate adaptive allocation, and the implications of emerging architectural frameworks for both design and evaluation. By situating the review findings within broader research trajectories, this section identifies areas of convergence, highlights points of divergence, and surfaces unresolved

Table 3 To analyse architectural contexts and evaluation practices supporting adaptive resource allocation

Author(s) & year	Architecture	Control framework	Evaluation platform	Evaluation maturity	Deployment readiness	Key strength	Limitation
Omheni et al. (2024)	O-RAN	Near-RT RIC	O-RAN emulator	High	High	Real-time feasibility	Limited large-scale validation
Aslan et al. (2024)	Virtualised O-RAN	O-Cloud	Simulation	Medium	Medium	Fairness improvement	Simulation-only evaluation
Ismail et al. (2025)	MEC	Edge orchestrator	Simulation	Medium	Medium	Latency reduction	Simplified assumptions
Firouzi & Rahmani (2024)	Edge AI IoT	Edge–cloud	Analytical + sim	Medium	Medium	QoE improvement	Limited dynamic realism
Mahboob et al. (2024)	NTN–6G	AI-assisted control	Simulation	Low	Low	Handles extreme mobility	Sparse real-world validation
Al-Hraishawi et al. (2024)	Digital Twin 6G	Closed-loop DT	Hybrid (DT + sim)	Very high	High	Risk-free testing	High complexity
Chen et al. (2023)	Integrated HetNets	Centralised DRL	Custom simulator	Medium	Low	SLA improvement	Lack of reproducibility
Wang et al. (2025)	IoV	Distributed MARL	Mobility simulator	Medium	Medium	Scalability	Limited real-world deployment
Zhang (2023)	Semantic 6G	Cross-layer	Conceptual + sim	Low	Low	Task-level efficiency	Early-stage concept
El-Hajj (2025)	Secure O-RAN	Federated control	Conceptual + sim	Low	Low	Security awareness	No experimental validation

challenges that will shape the evolution of adaptive resource allocation in 5G-Advanced and emerging 6G systems. In doing so, the discussion links empirical trends observed in the reviewed studies to ongoing theoretical and architectural debates within the wireless networking community.

Dynamic environment modelling as a foundational shift

A noteworthy takeaway from this review is that, while earlier generations of resource allocation research often assumed that traffic demand, interference, and user mobility were slow-varying processes that could be modelled as static or quasi-static, with tractable optimisation but limited real-world relevance, recent studies and surveys of 5G and 6G networks increasingly model non-stationarity as an intrinsic system characteristic, explicitly incorporating time-varying traffic patterns, user mobility, topology evolution, and service churn into problem formulations [17, 29], which aligns with broader trends identified in contemporary surveys of 5G and 6G networks, highlighting that densification, heterogeneity, and service diversification fundamentally alter network behaviour and invalidate static assumptions [2, 3]. Furthermore, studies in the context of vehicular, semantic, and non-terrestrial networking extend dynamic modelling from the physical layer to the task level, mission level, and context-dependent, reflecting the observation that resource demand is more and more driven by application semantics, perception objectives, and control requirements rather than raw data volume alone [18, 30,

34]. Taken together, the findings from this review suggest that dynamic modelling has become a necessity for effective adaptive resource allocation and a foundational change in how modern wireless systems are conceived and optimised.

Dominance of learning-based allocation and its implications

The review results indicate that learning-based methods, especially DRL and MARL, have become the de facto method for adaptive resource allocation in dynamic wireless environments due to their ability to solve sequential decision-making problems under uncertainty with complex system dynamics, partial observability, and difficulty in representing dynamics by closed-form models [12, 15]. DRL-based approaches can adjust allocation policies based on interactions with the environment without explicit re-optimisation at each time step. The literature also points to the limitations and risks of learning-based control, which include training instability, lack of convergence guarantees, and generalisation under distributional shifts, especially when learned policies are trained in simulation and then deployed in different real-world environments [25, 33]. The growing application of MARL in distributed scenarios (e.g., vehicular networks, dense multi-cell deployments, and multi-tenant environments) exacerbates the tensions between adaptability, scalability, and coordination overhead [19, 27]. These findings align with more general criticisms of learning-based optimisation without explicit mechanisms for safety, interpretability, and operational control.

Emergence of hybrid and constraint-aware approaches

A key insight emerging from the review is the growing prevalence of hybrid resource allocation frameworks that integrate learning-based techniques with optimisation constraints, architectural controls, or federated mechanisms. Rather than positioning learning as a wholesale replacement for classical optimisation, these approaches seek to embed domain knowledge, fairness constraints, or security requirements into adaptive decision-making processes [5, 9]. This hybridisation reflects an important maturation of the field, as researchers increasingly acknowledge the limitations of purely data-driven solutions in complex, safety–critical, and multi-tenant wireless systems. Studies situated within O-RAN environments illustrate how architectural constraints and policy enforcement mechanisms can guide learning-based xApps toward predictable, compliant, and auditable behaviour [20]. These developments align with emerging discussions around "trustworthy" or "responsible" AI in wireless networking, where performance gains must be balanced against robustness, fairness, and governance requirements. The rise of constraint-aware and policy-driven adaptive allocation frameworks suggests that future research will increasingly prioritise controlled adaptability, in which learning operates within clearly defined operational and regulatory boundaries.

Architectural expansion and cross-domain allocation

It also shows that adaptive resource allocation is becoming part of more complex, multi-domain architectural frameworks that move beyond radio access network to include multi-access edge computing (MEC), virtualized RANs, and non-terrestrial networks and are expanding the allocation problem to include radio, compute, storage, and control resources [10, 13] and cross-domain dependencies, such as the interaction between radio scheduling and edge workload placement, or between satellite mobility and

terrestrial handover mechanisms, which aligns with emerging 6G visions that emphasise end-to-end orchestration, AI-native operation, and tight coupling between communication and computation [7, 31]. While this review also shows that there is a gap between architectural sophistication (e.g., platforms such as O-RAN and digital twins introduce powerful control abstractions) and evaluation maturity (most studies rely on simulation-based validation, and hence their results may lack confidence in real-world deployability), there is a need for standardised testbeds, reproducible experimental frameworks, and incremental field trials to bridge the gap between architectural innovation and operational reality.

Evaluation practices and methodological limitations

The results also point to persistent limitations in evaluation practices across adaptive resource allocation studies: While performance metrics have expanded from throughput to include latency, reliability, fairness, energy efficiency, and task success, direct comparison across studies remains difficult because of heterogeneous assumptions, datasets, and simulation environments [6, 29]; this methodological fragmentation complicates meta-level assessment of progress and obscures fundamental trade-offs between competing approaches. Digital twin–assisted evaluation is a promising approach to address these challenges, allowing closed-loop testing under realistic dynamics without endangering live networks [4], but the literature also warns that digital twins themselves must be rigorously validated and continuously synchronized with physical systems to avoid propagating modelling biases [4]. Hence, enhanced evaluation rigor remains a crucial bottleneck in translating adaptive resource allocation research from controlled experimental settings to reliable, large-scale deployment.

Synthesis and implications for future research

Taken together, the discussion highlights a clear trajectory in adaptive resource allocation research toward dynamic environment modelling, learning-driven optimisation, and architecture-aware orchestration. At the same time, it exposes unresolved challenges related to stability, security, fairness, and empirical validation. These findings reinforce arguments in the literature that future progress will depend not only on algorithmic sophistication but also on holistic system design, governance mechanisms, and reproducible evaluation frameworks [2]; [3]. In this context, adaptive resource allocation should be understood as a socio-technical challenge that spans technical optimisation, architectural design, and operational governance. Addressing this complexity will be essential for realising the full potential of adaptive, intelligent wireless networks in 6G and beyond, where performance, trust, and accountability must coexist.

Conclusions and future work

This study has systematically reviewed adaptive resource allocation in dynamic wireless environments, highlighting a clear transition from static optimisation models to dynamic, learning-driven approaches capable of operating under non-stationary conditions. The findings show that deep reinforcement learning and multi-agent techniques dominate current research due to their ability to handle uncertainty and time-coupled decision processes. However, these approaches remain limited by issues of stability, generalisation, and deployment reliability, particularly in real-world scenarios. The review

further reveals a growing shift toward hybrid and constraint-aware frameworks, where learning-based methods are integrated with optimisation principles and architectural controls to improve robustness, fairness, and policy compliance. In addition, adaptive resource allocation is increasingly evolving into a cross-domain problem, requiring the joint optimisation of communication, computation, and storage resources, especially in edge-enabled and 6G systems. Despite these advances, a major limitation across existing studies is the overreliance on simulation-based evaluation, with insufficient use of real-world testbeds and reproducible experimental frameworks.

Future research should therefore focus on developing realistic dynamic models, advancing hybrid and trustworthy AI-driven allocation strategies, designing scalable distributed control mechanisms, and promoting reproducible evaluation through testbeds and digital-twin platforms. Addressing these challenges will be critical to enabling reliable, scalable, and deployable adaptive resource allocation in next-generation wireless networks.

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Author contributions
A. Clinton conceptualized the study, designed the review protocol, conducted the systematic search and analysis, and drafted the manuscript. S. Tweneboah-Koduah supervised the research, contributed to methodological design, and critically reviewed the manuscript. J. Owusu assisted with technical interpretation, literature synthesis, and manuscript revision. B. Kyiewu supported data extraction, analysis, and manuscript preparation. All authors reviewed and approved the final manuscript.

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Data availability
This study is a systematic review based on peer-reviewed literature published from 2020 onward. All data analysed in this study were obtained from publicly accessible scientific databases, including IEEE Xplore, Scopus, Web of Science, ACM Digital Library, ScienceDirect, and SpringerLink. Extracted data supporting the findings are available from the corresponding author upon reasonable request.

Declarations

Competing interests
The authors declare that they have no competing interests.

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